

Customer discovery

Part I · What are customers called where?

Sail · LakeCat · Pinax · MCP



One question across the company

“Sales says customer. Finance says account. Product says user. Where is each recorded?”

Find the table, field, definition, and responsible steward.

Three tables in the lakehouse

Department	Physical table	Fields
Sales	crm_customers	customer_id
Finance	billing_accounts	account_id, customer_ref
Product	product_users	user_id, customer_ref

Real Iceberg metadata and Parquet files · LakeCat catalog

What Pinax records

- A stable concept ID, definition, and alternative names
- Source references, steward, and review evidence
- Table bindings and stable field IDs at a registry revision

Concept approval and binding approval are separate.

Customer is an ambiguous name

Term	Meaning in this department
CRM customer	Organization with a commercial relationship
Billing account	Billing relationship; several per customer
Product user	Person with a login; may be a free user

Pinax returns the alternatives with their definitions.

Shared meaning at the field level

`crm_customers.customer_id`

`billing_accounts.customer_ref`

`product_users.customer_ref`

These three fields bind to **Enterprise customer identifier**.

`account_id` and `user_id` retain distinct concepts.

The agent asks through MCP

Discover tools

Search customer → account → user

Inspect every returned definition and binding

Find the enterprise customer reference

Console step 4 · Live signed metadata discovery

The answer names places and meanings

Sales: **customer** in `crm_customers`.

Finance: **account** in `billing_accounts`.

Product: **user** in `product_users`.

Follow-up: customer organizations, billing accounts, or individual users?

Maintain the mapping

1. Inventory and register tables and fields
2. Import draft concepts and bindings
3. Stewards review meanings and ambiguous names
4. Publish a revision and update consumer pins

Coverage means the authorized registered inventory.

Run the discovery demo

[Open demo.rust.ai](https://demo.rust.ai) · Iceberg

Choose the company table format in the header.

Four steps answer the customer naming question.

Part I ends here. Optional and Part II slides follow.

Optional · The whole stack

Table standards, agent access, and governed reads

The QueryGraph stack

Shared data, shared meaning, capable agents

Sail · LakeCat · Pinax · TypeSec · MCP · Marciana · Grust



A company asks a question

“Where is our customer data, and what can an analytics agent do with it?”

- Business teams use different words for the same concept
- Systems represent similar records with different columns
- Agents need meaning, structure, and usable operations

A lakehouse in Sail

Sail executes reads over Iceberg tables and Parquet records.

Field ID	Customer fixture column	Physical type
1	id	long, required
2	tenant	string, optional
3	private_email	string, optional

Optional · Retained customer read fixture

A catalog in LakeCat

The catalog identifies the physical table and the snapshot an operation will use.

- Table identity and schema
- Current snapshot and catalog state
- Planning and execution through the Sail adapter

One table, described physically by LakeCat and semantically by Pinax

A registry in Pinax

Standardize the shape of tables across a large company

Shared profile

Common foundation

Customers

Stable identities, typed fields, semantic terms

Employees

Common HR records and stewardship

Transactions

Consistent financial record structure

Optional · Generate enterprise standards

Shared meaning in the ontology

Account number → Customer identifier → rows.id

- Stable concepts, definitions, and aliases
- Reviewed bindings to registered tables and field IDs
- One maintained vocabulary for agents and exports

Optional · Ontology for the customer read fixture

Agents with TypeSec access

An identified agent discovers the tables and fields appropriate to its task.

- The analytics agent carries a TypeSec identity
- Pinax returns its usable table inventory
- Its requested operation carries a purpose and signature

Access is part of the normal workflow, alongside schema and meaning.

MCP is the agent's interface

Discover tools

- Search “account number” in Pinax
- Resolve the table and field
- Call `execute_pinax_scan`
- Receive rows and semantic metadata

Optional · MCP governed-read scenario

Plan and read the selected table

```
{"table":"rows","columns":["id"],  
  "purpose":"analytics","limit":10}
```

The plan identifies the current LakeCat snapshot.

Sail returns `[{"id":2}]`

Pinax's semantic exports preserve the selected field's meaning.

Optional · Inspect plan, then read selected rows

Part II · Graphs and workflows

Grust graph storage, semantic exports, and Marciana workflows

Grust connects the data

Datasets → files → fields → concepts

Live demonstration

Result

**Dataset
descriptions**

Two Dataverse fixtures

Property graph

34 nodes and 33 edges in Grust's Delta tables through Sail

Verification

Dataset node read back from Sail

Provenance

OpenLineage appended to the lakehouse

Part II · Build graph & run workflow

Marciana and semantic workflows

QueryGraph composes semantic and cognition capabilities into agent workflows.

- Navigator packages descriptions for discovery and reuse
- Specialists contribute domain summaries
- A supervisor coordinates and a synthesis agent combines
- TypeSec identities and lineage accompany the work

Part II · Semantic bundle and specialist workflow

Meaning travels with results

Surface	What an agent can reuse
Croissant	Record sets, fields, and bounded inline results
CDIF projection	Data elements linked to shared concepts
SKOS vocabulary	Labels, aliases, definitions, relationships
Lineage	The operations and inputs behind an answer

One central ontology feeds the semantic projections.

Run the whole-stack walkthrough

demo.rust.ai · Iceberg

Results stay on the left. Steps scroll on the right.

Click any component to see its role and a related operation.

Customer discovery · optional reads · graph workflows

Pinax

The agentic entrypoint to the QueryGraph stack

Shared table shapes

Shared business meaning

Discoverable operations

firstpair.org/books/pinax/

querygraph.ai/announcing-pinax/

